



Asymmetric effects of geopolitical risks on energy returns and volatility under different market conditions

Yun Qin^a, Kairong Hong^a, Jinyu Chen^{a,b,*}, Zitao Zhang^a

^a School of Business, Central South University, Changsha 410083, China

^b Institute of Metal Resources Strategy, Central South University, Changsha 410083, China

ARTICLE INFO

Article history:

Received 4 January 2020

Received in revised form 27 June 2020

Accepted 29 June 2020

Available online 7 July 2020

Keywords:

Geopolitical risks

Energy returns

Energy volatility

Different market conditions

Quantile regression

ABSTRACT

Based on daily data from 28 June 1990 to 31 October 2018, we investigate the asymmetric effects of geopolitical risks on energy (crude oil, gas and heating oil) returns and volatility under different market conditions by using a quantile regression model. The results show that geopolitical risks have significantly negative effects on crude oil returns in the bearish market and on heating oil returns in the normal and bullish markets but have no effects on gas returns. Meanwhile, the impacts of geopolitical risks on crude oil volatility are significantly positive in different market conditions, and the impacts on gas and heating oil volatility are mainly negative but not significant. In addition, we further divide geopolitical risks into geopolitical threats and geopolitical acts and find that geopolitical threats and acts have heterogeneous effects on energy returns and volatility.

© 2020 Elsevier B.V. All rights reserved.

1. Introduction

Energy is a vital strategic resource that plays a significant role in safeguarding national economic and social development (Wen et al., 2018; Wen et al., 2019). As the material basis for the survival and progress of human society, energy can be regarded as the source power for global economic growth (Chen and Wu, 2017; Chen and Zhu, 2019). In the process of a country's development, energy is considered the lifeblood of the national economy (Chen et al., 2019). The proportion of energy consumption in overall global consumption is increasing. Based on the BP Statistical Review of World Energy 2019, primary energy consumption increased by 2.9% in 2018, and its growth rate reached the highest level since 2010, at almost twice the average growth rate (1.5%) for the past decade. In view of the status of energy in economic development, energy price stability has become a critical issue that many countries pay close attention to. However, since energy is characterized by scarcity, important strategy, the spatial separation of supply and demand, and the low price elasticity of demand, energy prices are highly vulnerable to geopolitical risks (Su et al., 2019). Geopolitical risks refer to the risks related to terrorist acts, tensions and wars between countries (Caldara and Iacoviello, 2018), which are considered extreme risks, and their impact on the energy market is mainly reflected in expectation differences about the future energy supply and demand (Peng et al., 2018; Bouiyou et al., 2019; Liu et al., 2019; Demirel et al.,

2019). Meanwhile, because geopolitical risks are one of the most important determinants for investment decisions (Du and He, 2015; Caldara and Iacoviello, 2016). When extreme events related to geopolitical risks occur, they will cause panic among investors, lead to abnormal market fluctuations, and ultimately affect returns and volatility in the energy market (Antonakakis et al., 2017; Plakandaras et al., 2019; Mei et al., 2020). Therefore, it is necessary to examine the effects of geopolitical risks on the energy market.

Many experts have recently discussed the relationships between geopolitical risks and energy returns. Antonakakis et al. (2017) explored the relationship between oil-stock market associations and geopolitical risks. The results verified that the influence of geopolitical risks on oil returns was negative. Su et al. (2019) evaluated causal links between oil prices, geopolitical risks and financial liquidity and indicated that when geopolitical risks were high, oil prices were associated with financial liquidity in the time domain. Bouiyou et al. (2019) investigated the response of the oil market to geopolitical risks. The results showed that the impact of geopolitical acts was significantly positive, while the impact of geopolitical threats was non-significant. Plakandaras et al. (2019) investigated the dynamics between global geopolitical risks and oil prices and found that war-related geopolitical risks could predict oil returns correctly in the short run. Li et al. (2020) studied the links between crude oil prices and geopolitical risks. The results suggested that in the short run, there existed co-movement between the two variables at high frequencies, while in the long run, there existed no links between the two variables at low frequencies.

Other experts have focused on analyzing the effects of geopolitical risks on energy volatility. Wang and Yang (2018) investigated the

* Corresponding author at: School of Business, Central South University, Changsha 410083, China.

E-mail address: cjy19881001@163.com (J. Chen).

influence of geopolitical risks on oil price volatility during the shale revolution. The results indicated that the response of oil price volatility to geopolitical risks was higher with a shale production shock. Using the Generalized AutoRegressive Conditional Heteroskedasticity-Mixed Data Sampling (GARCH-MIDAS) model, Liu et al. (2019) examined the ability of geopolitical risks to predict oil volatility. The results verified that geopolitical risks caused fluctuations in the oil market. Demirer et al. (2019) aimed to determine whether geopolitical risks could forecast oil market volatility and found that geopolitical risks had significant effects on the volatility of the oil market. Using the Mixed Data Sampling (MIDAS) model, Mei et al. (2020) studied the influence of geopolitical risk uncertainty on oil futures price volatility. The results indicated that geopolitical risk uncertainty positively affected oil realized volatility in the short term. Asai et al. (2020) investigated the impact of geopolitical risks on the volatility of crude oil futures and found that geopolitical risks could promote forecast of the volatility of crude oil futures.

Based on the literature review above, we find that there are still some shortcomings in the previous studies. First, current studies have mainly examined the relationship between geopolitical risks and the crude oil market, but no study uses crude oil, natural gas and heating oil to represent energy and investigates the relationship between geopolitical risks and the energy market. Second, although many experts have studied the impacts of geopolitical risks on energy returns or energy volatility, only a few scholars have combined energy returns and energy volatility and examined the common influences on these two variables. Third, the current research mainly utilizes linear models such as Ordinary Least Square (OLS) and Vector AutoRegressive (VAR), which cannot reflect the nonlinear relationship between geopolitical risks and energy returns and volatility. The OLS model can only reveal the average relationship between the explanatory variable(s) and the dependent variable (Mosteller and Tukey, 1977). However, geopolitical risks have various impacts on energy returns and volatility in the bearish, normal, and bullish markets. Therefore, we should further explore the study area under different market conditions from a nonlinear perspective.

To summarize, we investigate the asymmetric effects of geopolitical risks on energy returns and volatility by utilizing a quantile regression model. Furthermore, considering the structural differences in geopolitical risks, we divide geopolitical risks index into geopolitical threats and geopolitical acts and explore the different effects of these two subindexes. In other words, the purpose of our study is to improve our understanding of how energy (crude oil, gas and heating oil) returns and volatility respond to geopolitical risks under different market conditions. The contributions of our study are as follows. First, due to the variety of energy resources, we select crude oil, gas and heating oil to represent energy (Reboredo and Uddin, 2016) and explore the heterogeneous influences of geopolitical risks on the crude oil, gas and heating oil markets. Furthermore, we investigate the effects on not only energy returns but also energy volatility. This means that we put energy returns and energy volatility into the same framework and explore the impacts of geopolitical risks on these two variables. In addition, taking the shortcomings of the OLS model into account, we use a quantile regression model to investigate the distributional heterogeneity of energy returns and volatility. The quantile regression model can capture the heterogeneous responses of the dependent variable to the explanatory variable(s) (Baur, 2013). Generally, we investigate the asymmetric effects of geopolitical risks on energy returns and volatility under different market conditions, including bearish (lower quantiles), normal (medium quantiles) and bullish (higher quantiles) markets.

The remaining three sections are structured as follows. Section 2 introduces the methodology and data. Section 3 discusses the empirical results and analysis. The conclusions and policy implications are presented in Section 4.

2. Methodology and data

2.1. Methodology

Relying on the quantile regression model, we explore the asymmetric effects of geopolitical risks on energy returns and volatility in this paper. The quantile regression model is based on the OLS model, which reveals the average links between the explanatory variable (s) and the dependent variable (Mosteller and Tukey, 1977). According to Nusair and Olson (2019), the OLS model is constructed as follows:

$$r_{it} = \alpha + \beta GPR_t + \delta D_t + \varepsilon_{it} \quad (1)$$

where r_{it} represents energy (crude oil, gas and heating oil) returns and volatility of energy i in time t , GPR_t refers to geopolitical risks, and $D_t(D_{t1}, \dots, D_{tm})'$ is the vector of n dummy variables. When $D_{it} = 1$, the observation is in the i^{th} period, and when $D_{it} = 0$, the observation is 0. ε_{it} denotes the random error term.

According to Caldara and Iacoviello (2018), there are four spikes in the geopolitical risks index, namely, the Gulf War, the 2003 Iraq invasion after 9/11, the 2014 Russia-Ukraine crisis, and the Paris terrorist attacks. To capture these spikes, four dummy variables are established and included in Eq. (1) to address the four major events that affect geopolitical risks. The dummy variables can be constructed as follows: D_1 explores the influence of the Gulf War; for the period from 2 August 1990 to 28 February 1991, its value is 1 and 0 otherwise. D_2 examines the effects of the 2003 Iraq invasion after 9/11, with a value of 1 from 11 September 2001 to 12 November 2001 and 0 otherwise. D_3 captures the impacts of the 2014 Russia-Ukraine crisis, and its value is 1 during the period from 20 March 2003 to 2 May 2003 and 0 otherwise. D_4 investigates the influence of the Paris terrorist attacks; when the period is from 21 November 2013 to 5 September 2014, the value is 1 and 0 otherwise.

However, Eq. (1) can only solve the problem of whether geopolitical risks have significant effects on energy returns and volatility but not the problem of whether geopolitical risks can affect energy returns and volatility under different market conditions. Based on Koenker and Bassett Jr (1978), You et al. (2017), and Demiralay and Kilincarslan (2019), we use quantile regression to address this issue. This approach can capture the nonlinear impacts of the explanatory variable(s) on the dependent variable in various quantiles (Nusair and Olson, 2019). The quantile regression is robust to heterogeneity, skewness and outlier observations for the dependent variable, so its result is more accurate (Xiao et al., 2018; Naifar et al., 2020). Generally, the conditional quantile regression model presented by Koenker and Bassett Jr (1978) is as follows:

$$Q_{y_i}(\tau|x) = \alpha(\tau) + x_i'\beta(\tau) \quad (2)$$

where $0 < \tau < 1$, and $Q_{y_i}(\tau|x)$ is the τ -th conditional quantile of y_i . x_i' refers to the explanatory variables. $\alpha(\tau)$ represents the unobserved effect. Following Xiao et al. (2019), $\beta(\tau)$ is the coefficient of the conditional quantile regression model and can be estimated as follows:

$$\hat{\beta}(\tau) = \arg \min_{\beta \in R^n} \sum_{i=1}^n \rho_{\tau}(y_i - x_i'\beta(\tau) - \alpha(\tau)) \quad (3)$$

where $\rho_{\tau} = u(\tau - I(u < 0))$ refers to the check function, and $I(\cdot)$ represents an indicator function ($u = y_i - x_i'\beta(\tau) - \alpha(\tau)$).

To investigate the effects of geopolitical risks on energy returns and volatility under different market conditions, we use the quantile regression model. Following Xiao et al. (2018), the OLS model corresponding to Eq. (1) can be transformed as follows:

$$Q_{r_{it}}(\tau|x) = \alpha(\tau) + \beta(\tau)GPR_t + \delta(\tau)D_t \quad (4)$$

Based on the quantile regression model in Eq. (4), following You et al. (2017), we use seven quantiles $\tau = (0.1, 0.2, 0.3, 0.5, 0.7, 0.8$ and

0.9) to capture the distribution of the energy market. The lower quantiles (0.1, 0.2 and 0.3) correspond to the bearish market, the medium quantile (0.5) represents the normal market, and the upper quantiles (0.7, 0.8 and 0.9) refer to the bullish market (Nusair and Olson, 2019; Zhu et al., 2019).

Following Andersen and Bollerslev (1998), Gong and Lin (2017), Gong and Lin (2018) and Demirer et al. (2018), we use the sum of squared daily energy returns to calculate the realized energy volatility. Energy volatility can be constructed as follows:

$$EV_t = \sum_{i=1}^M r_{it}^2 \quad (5)$$

where EV_t denotes energy volatility. r_{it} represents the daily M energy returns. $i = 1, \dots, M$ is the number of daily energy returns.

2.2. Data

The sample period selected in this paper is from 28 June 1990 to 31 October 2018. For energy returns, following Reboredo and Uddin (2016), gas and heating oil returns are measured by the first difference of the logarithmic form of futures prices collected at the New York Mercantile Exchange (NYMEX), and crude oil returns are measured by the first difference of the logarithmic form of West Texas Intermediate (WTI) futures prices. Meanwhile, energy volatility is measured by the sum of squared daily energy returns. These three types of energy futures prices are obtained from the Wind database. For geopolitical risks, we use the logarithmic form of the geopolitical risks index established by Caldara and Iacoviello (2018). Caldara and Iacoviello (2018) calculate the automatic text search results of 11 national and international newspaper electronic archives related to geopolitical risks and divide the geopolitical risks index into two subindexes, geopolitical threats and geopolitical acts. Geopolitical threats represent the direct influence of geopolitical events, and geopolitical acts represent the influence of pure geopolitical risks. The geopolitical risks index is obtained from <https://www.matteoiacoviello.com/gpr.htm>.

3. Empirical results and analysis

3.1. Unit root test

Based on the Augmented Dickey-Fuller (ADF) and Philips-Perron (PP) tests, we apply the unit root test to avoid spurious regression and examine stationarity (Nusair and Olson, 2019; Xiao et al., 2019; Chen et al., 2020). Table 1 shows that the original sequence of crude oil returns, gas returns, heating oil returns, crude oil volatility, gas volatility, heating oil volatility, geopolitical risks, geopolitical threats and geopolitical acts are stationary at the 1% significance level, so we select the original sequence of these variables as the data of our empirical study.

Table 1
The results of unit root test.

Variable	ADF test Trend and intercept	PP test Trend and intercept
Crude oil returns	−62.4886***	−85.7630***
Gas returns	−86.7672***	−87.0042***
Heating oil returns	−58.3204***	−123.6804***
Crude oil volatility	−23.0771***	−93.1975***
Gas volatility	−30.2778***	−92.0464***
Heating oil volatility	−17.2374***	−95.9456***
Geopolitical risks	−8.8449***	−100.5578***
Geopolitical threats	−11.6075***	−104.2248***
Geopolitical acts	−16.8433***	−105.6709***

Note: ***denotes significance at the 1% level.

3.2. The effects of geopolitical risks on energy returns and volatility

3.2.1. Geopolitical risks and energy returns

In this section, we explore the effects of geopolitical risks on energy returns. We first investigate the results of the OLS model in Table 2. The results show that geopolitical risks have a significantly negative impact on crude oil returns, indicating that high geopolitical risks result in low crude oil returns. The dummy variables are not significant for energy returns, suggesting that geopolitical risks have no influences on energy returns in the four major events.

In comparison to the OLS model, the quantile regression model examines the effects of geopolitical risks on energy returns in different quantiles. The results of the quantile regression model are presented in Table 2. Geopolitical risks have significantly negative impacts on crude oil returns at the lower quantiles of 0.1, 0.2 and 0.3, while they have a significantly positive impact at the upper quantile of 0.9. These results suggest that higher geopolitical risks lead to lower crude oil returns under the bearish market, while under the bullish market, geopolitical risks and crude oil returns show the same change. Geopolitical risks do not significantly affect gas returns, indicating that there exists no relationship between these two variables. The influences of geopolitical risks on heating oil returns are significantly negative at the medium quantile of 0.5 and at the upper quantiles of 0.7 and 0.8, indicating that increasing geopolitical risks result in decreasing heating oil returns in the normal and bullish markets. Furthermore, the majority of the four dummy variables are significant at different quantiles. Specifically, the effects of geopolitical risks on energy returns are significantly negative at the lower quantiles, while the effects are significantly positive at the upper quantiles during the periods of the Gulf War and the 2003 Iraq invasion after 9/11, except for gas returns in the Gulf War. During the 2014 Russia-Ukraine crisis, geopolitical risks minimally affect energy returns. In the period of the Paris terrorist attacks, geopolitical risks have significantly positive impacts on energy returns at the lower quantiles and have significantly negative impacts at the upper quantiles. These results verify that during these four major events, geopolitical risks have asymmetric impacts on energy returns in the bearish and bullish markets and have various influences in different major events.

According to the results for energy returns, we find that geopolitical risks have significantly negative impacts on crude oil returns at the lower quantiles and on heating oil returns at the medium and upper quantiles but have no impacts on gas returns in any quantile. The above results are similar to several studies in Antonakakis et al. (2017) and Plakandaras et al. (2019), which suggest that the influences of geopolitical risks on oil returns are negative. Moreover, during the four major events, the effects of geopolitical risks on energy returns are different and are asymmetric at the lower and upper quantiles. Since geopolitical risks are associated with terrorism, political tensions, war-like and other dangerous events, they can cause worldwide fluctuations and uncertainty (Kannadhasan and Das, 2019; Saint Akadiri et al., 2020; Tiwari et al., 2019). Therefore, the energy market is strongly affected by geopolitical risks (Antonakakis et al., 2017; Su et al., 2019). Generally, geopolitical risks will aggravate energy market turbulence and reduce energy returns (Liu et al., 2019). In addition, when major events occur, geopolitical risks will cause more intense turbulence in the energy market, resulting in asymmetric impacts under different market conditions.

Table 3 provides the quantile slope equality test to explore the statistical heterogeneity of the estimated coefficients of geopolitical risks. The null hypothesis is that the slopes are equal for every quantile. Accordingly, we investigate the results to determine whether the former quantile has the same slope as the latter quantile (e.g., $Q_{0.1} = Q_{0.2}$), whether the lower quantile has the same slope as the medium quantile ($Q_{0.1} = Q_{0.5}$), and whether the medium quantile has the same slope as the upper quantile ($Q_{0.5} = Q_{0.9}$). As seen in the results, the null hypothesis can be rejected in many quantiles for energy returns, implying that

Table 2
Quantile regression results for the effects of geopolitical risks on energy returns.

Variable	OLS	Q _{0.1}	Q _{0.2}	Q _{0.3}	Q _{0.5}	Q _{0.7}	Q _{0.8}	Q _{0.9}
Panel A. Crude oil returns								
Geopolitical risks	−0.0006*	−0.0021***	−0.0011***	−0.0009***	0.0001	0.0001	0.0004	0.0009*
D ₁	−0.0005	−0.0400***	−0.0222***	−0.0080**	0.0026	0.0153***	0.0221***	0.0311***
D ₂	−0.0054	−0.0069**	−0.0071*	−0.0045	−0.0072**	−0.0045	0.0053	0.0027
D ₃	−0.0043	−0.0128***	−0.0152**	−0.0127*	−0.0045	0.0074	0.0066**	0.0029
D ₄	−2.84E−05	0.0131***	0.0069***	0.0051***	−0.0005	−0.0049***	−0.0086***	−0.0124***
Constant	0.0026*	−0.0171***	−0.0108***	−0.0050***	1.12E−19	0.0090***	0.0144***	0.0218***
Panel B. Gas returns								
Geopolitical risks	0.0001	0.0010	0.0004	−0.0003	−0.0002	−0.0005	−0.0002	−0.0002
D ₁	−0.0004	0.0073*	0.0072***	0.0050*	−7.93E−05	−0.0076***	−0.0107***	−0.0176***
D ₂	0.0028	−0.0346***	−0.0285*	−0.0019	1.98E−05	0.0180***	0.0180*	0.0176***
D ₃	−0.0004	0.0056	0.0054	0.0025	0.0010	−0.0028	−0.0037	−0.0107***
D ₄	4.13E−05	0.0101***	0.0056**	0.0036*	−0.0002	−0.0024	−0.0067***	−0.0098***
Constant	−0.0004	−0.0406***	−0.0244***	−0.0127***	0.0006	0.0152***	0.0231***	0.0378***
Panel C. Heating oil returns								
Geopolitical risks	−0.0003	0.0004	0.0003	−0.0003	−0.0008**	−0.0012**	−0.0015**	0.0002
D ₁	0.0007	−0.0314***	−0.0229***	−0.0147***	−3.42E−05	0.0197***	0.0234***	0.0364***
D ₂	−0.0066	−0.0422***	−0.0279***	−0.0262	0.0001	0.0053	0.0071	0.0172*
D ₃	−0.0065	−0.0144***	−0.0249***	−0.0272**	−0.0080	0.0054	0.0080	0.0056
D ₄	−0.0004	0.0278***	0.0158***	0.0089***	0.0007	−0.0085***	−0.0159***	−0.0295***
Constant	0.0015	−0.0405***	−0.0239***	−0.0114***	0.0035**	0.0181***	0.0293***	0.0397***

Note: ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively.

Table 3
Quantile slope equality test for the effects of geopolitical risks on energy returns.

Variable	Q _{0.1} = Q _{0.2}	Q _{0.2} = Q _{0.3}	Q _{0.3} = Q _{0.5}	Q _{0.5} = Q _{0.7}	Q _{0.7} = Q _{0.8}	Q _{0.8} = Q _{0.9}	Q _{0.1} = Q _{0.5}	Q _{0.5} = Q _{0.9}
Panel A. Crude oil returns								
Geopolitical risks	−0.0010***	−0.0002	−0.0010***	−8.33E−06	−0.0002	−0.0005	−0.0022***	−0.0008*
D ₁	−0.0178	−0.0142***	−0.0106***	−0.0127***	−0.0068*	−0.0091**	−0.0426***	−0.0285***
D ₂	0.0002	−0.0025	0.0027	−0.0027	−0.0098**	0.0025	0.0003	−0.0010
D ₃	0.0024	−0.0025	−0.0082	−0.0119	0.0008	0.0037	−0.0083	−0.0074
D ₄	0.0061***	0.0018**	0.0057***	0.0043***	0.0037***	0.0038**	0.0138***	0.0119***
Panel B. Gas returns								
Geopolitical risks	0.0006	0.0006*	−8.31E−05	0.0003	−0.0002	−8.21E−06	0.0012	5.07E−05
D ₁	0.0002	0.0022	0.0055***	0.0075***	0.0031**	0.0069***	0.0079**	0.0175***
D ₂	−0.0061	−0.0266**	−0.0019	−0.0180**	2.66E−05	0.0004	−0.0346***	−0.0175*
D ₃	0.0002	0.0029	0.0014	0.0038	0.0009	0.0070	0.0046	0.0117***
D ₄	0.0046**	0.0020	0.0039**	0.0022	0.0043***	0.0030	0.0104***	0.0096***
Panel C. Heating oil returns								
Geopolitical risks	0.0002	0.0006	0.0005	0.0004	0.0003	−0.0017**	0.0013	−0.0010
D ₁	−0.0085	−0.0082**	−0.0138***	−0.0197***	−0.0037	−0.0130***	−0.0305***	−0.0365***
D ₂	−0.0143***	−0.0016	−0.0264**	−0.0052	−0.0018	−0.0100	−0.0424***	−0.0170*
D ₃	0.0104**	0.0023	−0.0192**	−0.0134	−0.0027	0.0025	−0.0064	−0.0136
D ₄	0.0120***	0.0069***	0.0081***	0.0091***	0.0075***	0.0135***	0.0271***	0.0301***

Note: ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively.

the estimated coefficients of geopolitical risks vary with the various quantiles.

3.2.2. Geopolitical risks and energy volatility

We explore the impacts of geopolitical risks on energy volatility in this section. First, we display the results of the OLS model in Table 4. Geopolitical risks have a significantly positive effect on crude oil volatility, while they have a significantly negative effect on gas volatility. These results imply that greater geopolitical risks cause greater crude oil volatility and lead to lower gas volatility. In addition, most of the dummy variables are significant in energy volatility, indicating that geopolitical risks have significant impacts on energy volatility in the four major events.

For the results of the quantile regression model, Table 4 shows that geopolitical risks significantly positively affect crude oil volatility across the quantiles except for 0.1, suggesting that higher geopolitical risks lead to higher crude oil volatility under different market conditions. Geopolitical risks exert significantly negative effects on gas volatility at the upper quantile of 0.9 and on heating oil volatility at the medium quantile of 0.5. These results indicate that the effects on gas and heating

oil volatility are only significantly negative in the bullish and normal markets, respectively. Moreover, the majority of the four dummy variables are significant at various quantiles. In particular, the impacts of geopolitical risks on energy volatility are significantly positive at different quantiles during the Gulf War, the 2003 Iraq invasion after 9/11 and the 2014 Russia-Ukraine crisis, with the exception of gas volatility in the Gulf War and the 2014 Russia-Ukraine crisis. Geopolitical risks significantly negatively affect energy volatility at different quantiles during the period of the Paris terrorist attacks. These results verify that geopolitical risks have different effects on energy volatility in these four major events.

From the above results, we conclude that geopolitical risks have significantly positive impacts on crude oil volatility at different quantiles, which is consistent with previous studies related to the effects of geopolitical risks on oil volatility (Demirer et al., 2019; Liu et al., 2019; Mei et al., 2020). Relatively speaking, the impacts on gas and heating oil volatility are mainly negative but not significant. Furthermore, the effects of geopolitical risks on energy volatility are different in the four major events. Geopolitical risks can increase energy market fluctuations and reduce economic stability, thereby intensifying energy volatility (Jin,

Table 4

Quantile regression results for the effects of geopolitical risks on energy volatility.

Variable	OLS	Q _{0.1}	Q _{0.2}	Q _{0.3}	Q _{0.5}	Q _{0.7}	Q _{0.8}	Q _{0.9}
Panel A. Crude oil volatility								
Geopolitical risks	7.41E-05**	9.83E-07	2.63E-06*	4.85E-06***	1.62E-05***	3.58E-05***	6.75E-05***	9.45E-05**
D ₁	0.0032***	2.18E-05	0.0001**	0.0003***	0.0007***	0.0018***	0.0032***	0.0075***
D ₂	0.0006*	3.61E-06	5.91E-06	2.11E-05	8.38E-05	0.0005	0.0005***	0.0007***
D ₃	0.0007*	4.16E-05	8.59E-05	0.0002*	0.0004***	0.0006**	0.0011***	0.0020***
D ₄	-0.0004**	-4.12E-06	-1.44E-05***	-3.40E-05***	-0.0001***	-0.0003***	-0.0005***	-0.0010***
Constant	0.0002	6.74E-07	8.74E-06	2.50E-05***	7.98E-05***	0.0002***	0.0004***	0.0009***
Panel B. Gas volatility								
Geopolitical risks	-9.63E-05**	5.90E-07	1.91E-07	-3.06E-07	-1.06E-05	-2.04E-05	-3.77E-05	-0.0001**
D ₁	0.0001	-9.87E-06	-3.70E-05***	-7.75E-05***	-0.0002***	-0.0005***	-0.0008***	-0.0013***
D ₂	0.0024***	4.96E-05	0.0002**	0.0002*	0.0009*	0.0021***	0.0035***	0.0050***
D ₃	-0.0005	-8.98E-06	-2.59E-05	-6.89E-05	-0.0001	-0.0003	-0.0005**	-0.0012***
D ₄	-0.0002	-6.64E-06	-2.28E-05	-4.23E-05***	-0.0001***	-0.0003***	-0.0007***	-0.0009***
Constant	0.0015***	7.85E-06	4.27E-05***	9.89E-05***	0.0004***	0.0009***	0.0015***	0.0032***
Panel C. Heating oil volatility								
Geopolitical risks	-1.15E-06	1.22E-07	-2.98E-06	-5.20E-06	-2.34E-05**	-3.16E-05	-1.96E-05	9.14E-07
D ₁	0.0027***	5.65E-05*	0.0001***	0.0004***	0.0011***	0.0023***	0.0041***	0.0052***
D ₂	0.0017***	4.82E-05	0.0001**	0.0001*	0.0006*	0.0018***	0.0047***	0.0046***
D ₃	0.0009*	6.36E-05	0.0002	0.0004**	0.0006*	0.0014***	0.0012***	0.0033***
D ₄	-0.0010***	-7.31E-06	-2.94E-05***	-7.09E-05***	-0.0003***	-0.0008***	-0.0014***	-0.0028***
Constant	0.0011***	7.93E-06	4.72E-05***	0.0001***	0.0004***	0.0010***	0.0016***	0.0030***

Note: ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively.

2015). Compared with other energy sources, oil is the most important raw material in modern industrial society. Additionally, during the different major events, geopolitical risks have diverse influences on the energy market, so the impacts on energy volatility vary.

Table 5 presents the quantile slope equality test to investigate the statistical heterogeneity of the estimated coefficients of geopolitical risks. The results suggest that the null hypothesis can be rejected at many quantiles for energy volatility. These results indicate that the estimated coefficients of geopolitical risks are different at various quantiles.

3.3. The heterogeneous effects of geopolitical threats, acts on energy returns and volatility

3.3.1. Geopolitical threats, acts and energy returns

We divide geopolitical risks into geopolitical threats and geopolitical acts and compare the different impacts of the two subindexes on energy returns in Tables 6 and 7. The results of the OLS model show that geopolitical threats have no influences on energy returns, while geopolitical acts have a significantly negative influence on gas returns. These results imply that geopolitical threats and acts have various effects on energy returns.

For the results of the quantile regression model, geopolitical threats and acts both significantly negatively affect crude oil returns at the lower quantiles of 0.1, 0.2 and 0.3. These indicate that greater geopolitical threats and acts lead to lower crude oil returns in the bearish market. The results are consistent with the results for geopolitical risks in Table 2, but the degrees of the influences are different. In regard to gas returns, the influence of geopolitical threats is significantly positive at the lower quantile of 0.1, while the influences of geopolitical acts are significantly negative at the lower quantiles of 0.1, 0.2 and 0.3, which are distinct from the results for geopolitical risks. Regarding heating oil returns, geopolitical threats have significantly negative effects at the medium quantile of 0.5 and at the upper quantiles of 0.7 and 0.8, while geopolitical acts have significantly negative impacts at the lower quantiles of 0.1, 0.2 and 0.3 but have a significantly positive impact at the upper quantile of 0.7. The results related to the effects of geopolitical threats are similar to the results for geopolitical risks but with differences in the degrees of the effects.

We put geopolitical threats and geopolitical acts into the same model and display the results in Fig. 1. Geopolitical threats and acts have different impacts on the three types of energy returns at various

Table 5

Quantile slope equality test for the effects of geopolitical risks on energy volatility.

Variable	Q _{0.1} = Q _{0.2}	Q _{0.2} = Q _{0.3}	Q _{0.3} = Q _{0.5}	Q _{0.5} = Q _{0.7}	Q _{0.7} = Q _{0.8}	Q _{0.8} = Q _{0.9}	Q _{0.1} = Q _{0.5}	Q _{0.5} = Q _{0.9}
Panel A. Crude oil volatility								
Geopolitical risks	-1.65E-06	-2.22E-06*	-1.13E-05***	-1.96E-05***	-3.17E-05**	-2.70E-05	-1.52E-05***	-7.83E-05**
D ₁	-0.0001**	-0.0002***	-0.0004*	-0.0010**	-0.0014***	-0.0044***	-0.0007***	-0.0067***
D ₂	-2.30E-06	-1.52E-05	-6.27E-05	-0.0004	-4.76E-05	-0.0002	-8.02E-05	-0.0006**
D ₃	-4.43E-05	-6.50E-05	-0.0002***	-0.0002	-0.0005***	-0.0010***	-0.0003***	-0.0017***
D ₄	1.02E-05**	1.96E-05***	7.84E-05***	0.0002***	0.0002***	0.0005***	0.0001***	0.0008***
Panel B. Gas volatility								
Geopolitical risks	3.99E-07	4.97E-07	1.03E-05	9.86E-06	1.73E-05	9.61E-05**	1.12E-05	0.0001**
D ₁	2.72E-05**	4.04E-05***	0.0001***	0.0003***	0.0003***	0.0004	0.0002***	0.0011***
D ₂	-0.0001**	-9.81E-06	-0.0007*	-0.0013**	-0.0014***	-0.0015***	-0.0008**	-0.0041***
D ₃	1.69E-05	4.31E-05	5.31E-05	0.0002	0.0002	0.0006***	0.0001	0.0010***
D ₄	1.61E-05	1.95E-05*	7.05E-05***	0.0002***	0.0003***	0.0002	0.0001***	0.0008**
Panel C. Heating oil volatility								
Geopolitical risks	3.10E-06	2.22E-06	1.82E-05**	8.15E-06	-1.20E-05	-2.05E-05	2.36E-05**	-2.43E-05
D ₁	-8.81E-05**	-0.0002***	-0.0008***	-0.0010***	-0.0018***	-0.0011	-0.0011***	-0.0039***
D ₂	-6.76E-05	-6.50E-06	-0.0004*	-0.0012***	-0.0029***	9.31E-05	-0.0005*	-0.0040***
D ₃	-0.0001	-0.0002**	-0.0002	-0.0008**	0.0002	-0.0021***	-0.0005**	-0.0027***
D ₄	2.21E-05***	4.15E-05***	0.0002***	0.0006***	0.0006***	0.0013***	0.0003***	0.0025***

Note: ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively.

Table 6
Quantile regression results for the effects of geopolitical threats on energy returns.

Variable	OLS	Q _{0.1}	Q _{0.2}	Q _{0.3}	Q _{0.5}	Q _{0.7}	Q _{0.8}	Q _{0.9}
Panel A. Crude oil returns								
Geopolitical threats	−0.0005	−0.0017***	−0.0009***	−0.0008***	7.36E-05	4.53E-05	0.0003	0.0008*
D ₁	−0.0006	−0.0405***	−0.0223***	−0.0079**	0.0026	0.0153***	0.0221***	0.0312***
D ₂	−0.0056	−0.0076**	−0.0076*	−0.0047	−0.0071**	−0.0045	0.0054	0.0030
D ₃	−0.0045	−0.0137***	−0.0153**	−0.0134*	−0.0044	0.0076	0.0068**	0.0035
D ₄	−3.25E-05	0.0134***	0.0069***	0.0051***	−0.0005	−0.0049***	−0.0085***	−0.0125***
Constant	0.0021*	−0.0191***	−0.0116***	−0.0056***	0.0002	0.0094***	0.0147***	0.0222***
Panel B. Gas returns								
Geopolitical threats	0.0003	0.0014**	0.0004	−0.0001	−0.0001	−0.0004	−0.0002	−0.0002
D ₁	−0.0006	0.0073*	0.0070**	0.0049*	−0.0002	−0.0076***	−0.0107***	−0.0176***
D ₂	0.0025	−0.0349***	−0.0286*	−0.0021	−8.92E-05	0.0179***	0.0178*	0.0175***
D ₃	−0.0007	0.0054	0.0054	0.0022	0.0009	−0.0031	−0.0038	−0.0108***
D ₄	−8.07E-05	0.0098***	0.0055**	0.0036*	−0.0003	−0.0024	−0.0067***	−0.0098***
Constant	−0.0012	−0.0421***	−0.0243***	−0.0132***	0.0004	0.0149***	0.0229***	0.0378***
Panel C. Heating oil returns								
Geopolitical threats	−2.83E-05	0.0011	0.0005	−0.0002	−0.0007**	−0.0012***	−0.0014**	0.0005
D ₁	0.0005	−0.0322***	−0.0230***	−0.0148***	−0.0002	0.0199***	0.0232***	0.0357***
D ₂	−0.0071	−0.0431***	−0.0276***	−0.0265	−0.0002	0.0054	0.0069	0.0164*
D ₃	−0.0070	−0.0160***	−0.0249***	−0.0275***	−0.0082	0.0052	0.0072	0.0048
D ₄	−0.0005	0.0276***	0.0157***	0.0092***	0.0006	−0.0083***	−0.0158***	−0.0295***
Constant	0.0004	−0.0432***	−0.0248***	−0.0120***	0.0029**	0.0182***	0.0290***	0.0385***

Note: ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively.

quantiles. Moreover, the results associated with the influences of geopolitical threats on crude oil and heating oil returns are consistent with the results for geopolitical risks in Table 2 but have different degrees of the influences.

Comparing geopolitical threats and geopolitical acts, we find that geopolitical threats have significantly different impacts on the three types of energy returns at various quantiles, while geopolitical acts have significantly negative impacts on energy returns at the lower quantiles. Generally, geopolitical threats and acts have diverse effects on energy returns. A plausible explanation is that geopolitical threats are mainly related to military or nuclear tensions and war or terrorist threats, while geopolitical acts are particularly concerned with terrorist attacks or the beginning of the war (Caldara and Iacoviello, 2018; Bouiyou et al., 2019; Demiralay and Kilincarslan, 2019). Therefore, geopolitical threats and geopolitical acts affect the three types of energy returns in different ways.

3.3.2. Geopolitical threats, acts and energy volatility

We capture the different effects of geopolitical threats and geopolitical acts on energy volatility and present the results in Tables 8 and 9. The results of the OLS model show that geopolitical threats significantly positively affect crude oil volatility, while they significantly negatively affect gas volatility. The impacts of geopolitical acts on crude oil and heating oil volatility are significantly positive. These results indicate that geopolitical threats and acts have various effects on energy volatility.

The results for the quantile regression model suggest that geopolitical threats and acts have significantly positive influences on crude oil volatility across all quantiles except for 0.1, implying that increasing geopolitical threats and acts cause increasing crude oil volatility under different market conditions. Regarding gas volatility, the effect of geopolitical threats is significantly negative at the upper quantile of 0.9, and the effect of geopolitical acts is significantly positive at the medium

Table 7
Quantile regression results for the effects of geopolitical acts on energy returns.

Variable	OLS	Q _{0.1}	Q _{0.2}	Q _{0.3}	Q _{0.5}	Q _{0.7}	Q _{0.8}	Q _{0.9}
Panel A. Crude oil returns								
Geopolitical acts	−0.0002	−0.0004*	−0.0004**	−0.0002*	1.75E-05	0.0002	6.59E-05	0.0002
D ₁	−0.0010	−0.0400***	−0.0233***	−0.0088***	0.0027	0.0158***	0.0224***	0.0324***
D ₂	−0.0057	−0.0096***	−0.0081*	−0.0052	−0.0070**	−0.0048	0.0056	0.0036
D ₃	−0.0045	−0.0162***	−0.0156**	−0.0136*	−0.0044	0.0071	0.0072**	0.0035
D ₄	−0.0003	0.0133***	0.0069***	0.0042***	−0.0004	−0.0049***	−0.0084***	−0.0122***
Constant	0.0008*	−0.0242***	−0.0142***	−0.0080***	0.0005	0.0091***	0.0157***	0.0248***
Panel B. Gas returns								
Geopolitical acts	−0.0003*	−0.0006*	−0.0004**	−0.0005**	−0.0003	−0.0001	−0.0002	−0.0003
D ₁	−9.64E-05	0.0094**	0.0078***	0.0060***	−0.0005	−0.0079***	−0.0113***	−0.0177***
D ₂	0.0042	−0.0305***	−0.0262*	−0.0007	0.0006	0.0180***	0.0184*	0.0184***
D ₃	0.0009	0.0099	0.0079	0.0041	0.0015	−0.0032	−0.0035	−0.0100**
D ₄	4.63E-05	0.0102***	0.0059***	0.0038*	−0.0005	−0.0028	−0.0069***	−0.0096***
Constant	0.0011*	−0.0351***	−0.0216***	−0.0124***	0.0005	0.0137***	0.0227***	0.0379***
Panel C. Heating oil returns								
Geopolitical acts	−0.0003	−0.0012***	−0.0008***	−0.0007***	−0.0002	0.0004**	0.0002	0.0003
D ₁	0.0006	−0.0296***	−0.0228***	−0.0139**	−0.0008	0.0183***	0.0211***	0.0370***
D ₂	−0.0063	−0.0373***	−0.0257***	−0.0242	−0.0006	0.0021	0.0040	0.0164*
D ₃	−0.0061	−0.0085**	−0.0216***	−0.0251***	−0.0082	0.0032	0.0047	0.0045
D ₄	−0.0006	0.0261***	0.0157***	0.0092***	0.0004	−0.0092***	−0.0167***	−0.0297***
Constant	0.0010	−0.0353***	−0.0206***	−0.0109***	0.0005	0.0122***	0.0228***	0.0397***

Note: ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively.

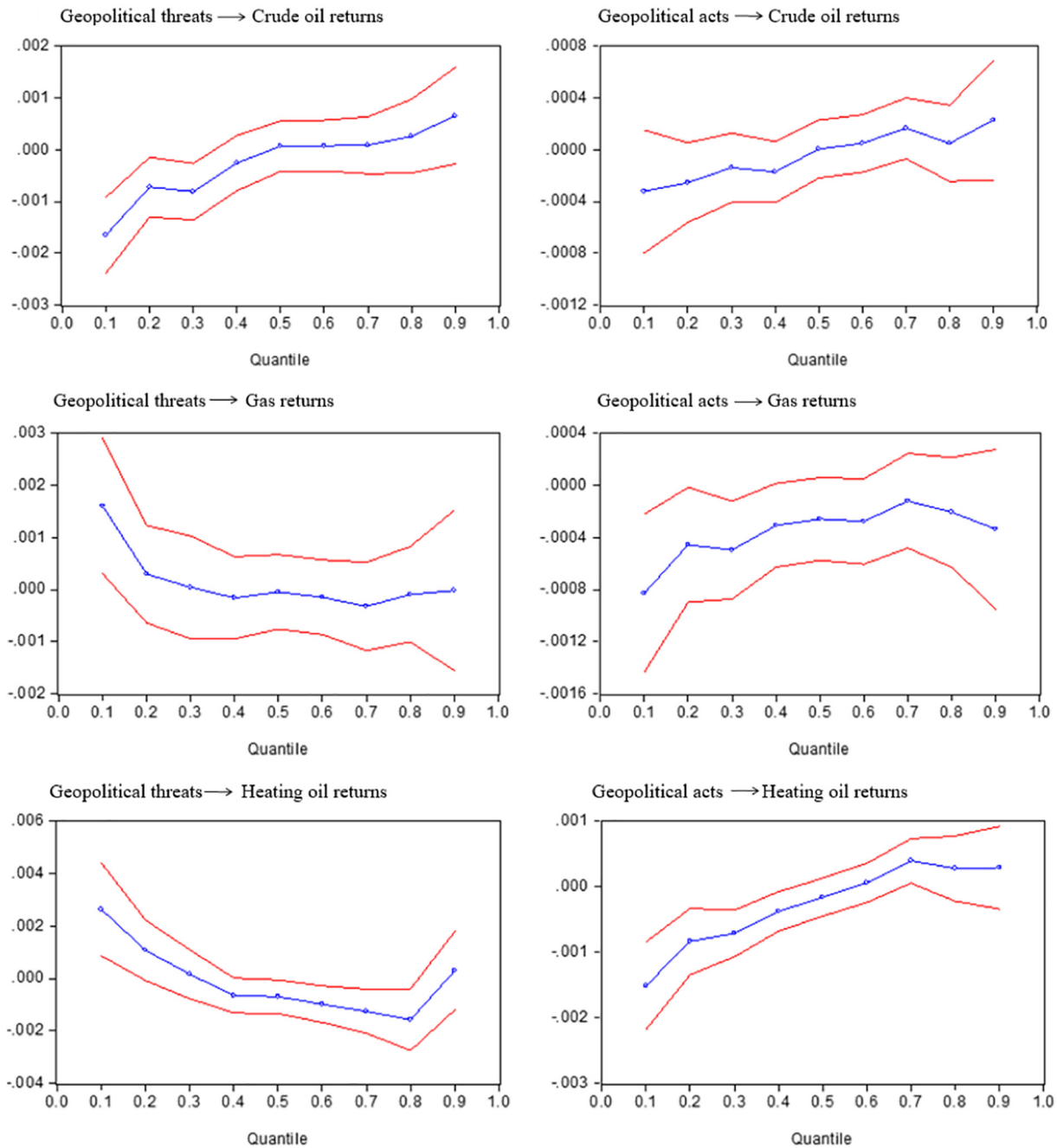


Fig. 1. Quantile regression results with 95% confidence intervals for the effects of geopolitical threats and acts on energy returns.

quantile of 0.5, so the effects are negligible. These results are similar to the results for geopolitical risks in Table 4, and the degrees of the influences are various. For heating oil volatility, the influences of geopolitical threats are significantly negative at the lower quantile of 0.3, at the medium quantile of 0.5 and at the upper quantile of 0.7, and the influences of geopolitical acts are significantly positive across the quantiles except for 0.1, 0.2 and 0.9, and these results are distinct from the results for geopolitical risks.

Putting geopolitical threats and geopolitical acts into the same model, we present the results in Fig. 2. Similar to the results in Table 4, the impacts of geopolitical threats and acts on crude oil volatility are significantly positive at different quantiles, and the influences on gas volatility are minimal. Furthermore, geopolitical

threats and acts have various effects on heating oil volatility at different quantiles.

Comparing the geopolitical threats and geopolitical acts, we can draw the following conclusions. Geopolitical threats and acts have significantly positive impacts on energy volatility at different quantiles, except for the effects of geopolitical threats on gas and heating oil volatility, which are significantly negative. Geopolitical threats and acts are geopolitical risks associated with terrorist acts, military tensions, war and nuclear threats (Caldara and Iacoviello, 2016; Demiralay and Kilincarslan, 2019). The higher the geopolitical risks, the higher the uncertainty of the energy market, and the greater the risks in the energy market, which will bring increasing energy volatility (Demirer et al., 2018).

Table 8
Quantile regression results for the effects of geopolitical threats on energy volatility.

Variable	OLS	Q _{0.1}	Q _{0.2}	Q _{0.3}	Q _{0.5}	Q _{0.7}	Q _{0.8}	Q _{0.9}
Panel A. Crude oil volatility								
Geopolitical threats	6.28E-05**	6.29E-07	2.13E-06	3.76E-06**	1.21E-05***	2.99E-05***	6.11E-05***	8.95E-05***
D ₁	0.0032***	2.22E-05	0.0001**	0.0003***	0.0007**	0.0018***	0.0031***	0.0075***
D ₂	0.0006*	4.29E-06	7.06E-06	2.22E-05	8.94E-05	0.0005	0.0005***	0.0007***
D ₃	0.0008*	4.25E-05	8.67E-05	0.0002**	0.0004***	0.0006**	0.0011***	0.0021***
D ₄	-0.0004***	-4.11E-06	-1.41E-05***	-3.47E-05***	-0.0001***	-0.0003***	-0.0005***	-0.0010***
Constant	0.0003**	2.12E-06	1.06E-05**	2.98E-05***	9.68E-05***	0.0003***	0.0004***	0.0009***
Panel B. Gas volatility								
Geopolitical threats	-0.0001***	1.18E-06	6.44E-07	2.52E-07	-1.50E-05	-1.91E-05	-5.14E-05	-0.0002***
D ₁	0.0002	-1.03E-05	-3.77E-05***	-7.77E-05***	-0.0002***	-0.0005***	-0.0008***	-0.0012***
D ₂	0.0024***	4.81E-05	0.0002**	0.0002*	0.0009**	0.0022***	0.0035***	0.0050***
D ₃	-0.0004	-9.90E-06	-2.68E-05	-7.00E-05*	-0.0001	-0.0003	-0.0005**	-0.0011***
D ₄	-0.0002	-7.47E-06	-2.33E-05	-4.22E-05**	-0.0001***	-0.0003***	-0.0006***	-0.0008**
Constant	0.0017***	5.60E-06	4.11E-05***	9.64E-05***	0.0004***	0.0009***	0.0016***	0.0033***
Panel C. Heating oil volatility								
Geopolitical threats	-2.22E-05	4.57E-09	-2.96E-06	-6.70E-06*	-2.64E-05***	-3.86E-05*	-2.56E-05	-3.14E-05
D ₁	0.0027***	5.66E-05*	0.0002***	0.0004***	0.0011***	0.0023***	0.0041***	0.0052***
D ₂	0.0017***	4.85E-05	0.0001**	0.0001*	0.0006**	0.0018***	0.0047***	0.0047***
D ₃	0.0009**	6.37E-05	0.0002	0.0004**	0.0006**	0.0014***	0.0012***	0.0033***
D ₄	-0.0010***	-7.32E-06	-2.95E-05***	-7.06E-05***	-0.0003***	-0.0008***	-0.0014***	-0.0027***
Constant	0.0012***	8.41E-06	4.72E-05***	0.0001***	0.0004***	0.0011***	0.0017***	0.0031***

Note: ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively.

4. Conclusions and policy implications

We investigate the asymmetric effects of geopolitical risks on energy (crude oil, gas and heating oil) returns and volatility over the period from 28 June 1990 to 31 October 2018 by applying a quantile regression model. Moreover, we divide the geopolitical risks index into geopolitical threats and geopolitical acts and explore whether the two subindexes have different effects on energy returns and volatility. The conclusions are as follows.

Using the OLS model, the results show that geopolitical risks have a significantly negative impact on crude oil returns, a significantly positive impact on crude oil volatility and a significantly negative impact on gas volatility. These indicate that high geopolitical risks lead to low crude oil returns, high crude oil volatility and low gas volatility. For the results of the dummy variables, there exists no relationship between geopolitical risks and energy returns, but geopolitical risks significantly affect energy volatility in the four major events.

Based on the quantile regression model, we find that geopolitical risks significantly negatively affect crude oil returns at the lower quantiles and affect heating oil returns at the medium and upper quantiles but do not affect gas returns in various quantiles. These imply that higher geopolitical risks cause lower crude oil returns in the bearish market and result in lower heating oil returns in the normal and bullish markets but do not cause gas returns changes in different market conditions. Geopolitical risks have significantly positive effects on crude oil volatility in various quantiles, meaning that higher geopolitical risks lead to higher crude oil volatility under different market conditions. However, the effects on gas and heating oil volatility are mainly negative but not significant. Considering the dummy variables, the results indicate that the influences of geopolitical risks on energy returns are asymmetric in the bearish and bullish markets, and the influences on energy volatility are various during the four major events.

The impacts of geopolitical threats on the three types of energy returns are diverse at various quantiles, and the impacts of geopolitical acts on energy returns are significantly negative at the lower quantiles.

Table 9
Quantile regression results for the effects of geopolitical acts on energy volatility.

Variable	OLS	Q _{0.1}	Q _{0.2}	Q _{0.3}	Q _{0.5}	Q _{0.7}	Q _{0.8}	Q _{0.9}
Panel A. Crude oil volatility								
Geopolitical acts	2.09E-05*	1.88E-07	1.23E-06**	1.51E-06**	5.27E-06***	8.71E-06**	1.69E-05**	2.42E-05
D ₁	0.0033***	2.28E-05	0.0001**	0.0003***	0.0007**	0.0018***	0.0032***	0.0076***
D ₂	0.0007*	5.28E-06	3.77E-06	2.29E-05	9.54E-05	0.0005	0.0006**	0.0008***
D ₃	0.0008*	4.29E-05	8.65E-05	0.0002**	0.0004***	0.0006**	0.0012***	0.0021***
D ₄	-0.0004**	-3.47E-06	-1.23E-05**	-3.20E-05***	-0.0001***	-0.0003***	-0.0005***	-0.0009***
Constant	0.0004***	3.95E-06**	1.57E-05***	4.09E-05***	0.0001***	0.0004***	0.0006***	0.0012***
Panel B. Gas volatility								
Geopolitical acts	4.70E-06	-2.01E-07	5.69E-07	1.23E-06	6.52E-06*	8.46E-06	1.05E-05	1.37E-05
D ₁	1.09E-05	-9.02E-06	-3.76E-05***	-7.85E-05***	-0.0002***	-0.0005***	-0.0009***	-0.0014***
D ₂	0.0022***	5.17E-05	0.0002**	0.0002*	0.0008**	0.0021***	0.0034***	0.0047***
D ₃	-0.0007	-7.36E-06	-2.80E-05	-7.43E-05*	-0.0001*	-0.0003*	-0.0006**	-0.0015***
D ₄	-0.0003	-5.69E-06	-2.26E-05	-4.24E-05***	-0.0001***	-0.0003***	-0.0007***	-0.0009***
Constant	0.0011***	1.08E-05***	4.25E-05***	9.41E-05***	0.0003***	0.0008***	0.0013***	0.0026***
Panel C. Heating oil volatility								
Geopolitical acts	4.17E-05***	5.52E-07	2.09E-06	4.14E-06***	1.67E-05***	3.32E-05***	4.76E-05***	4.51E-05
D ₁	0.0026***	5.79E-05**	0.0001***	0.0004***	0.0011***	0.0021***	0.0041***	0.0051***
D ₂	0.0015***	4.64E-05	0.0001*	9.35E-05	0.0005	0.0016***	0.0045***	0.0044***
D ₃	0.0007	6.22E-05	0.0002	0.0004**	0.0005*	0.0012***	0.0009**	0.0031***
D ₄	-0.0010***	-7.07E-06	-2.94E-05***	-7.37E-05***	-0.0003***	-0.0008***	-0.0014***	-0.0027***
Constant	0.0010***	7.07E-06**	2.94E-05***	7.37E-05***	0.0003***	0.0008***	0.0014***	0.0028***

Note: ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively.

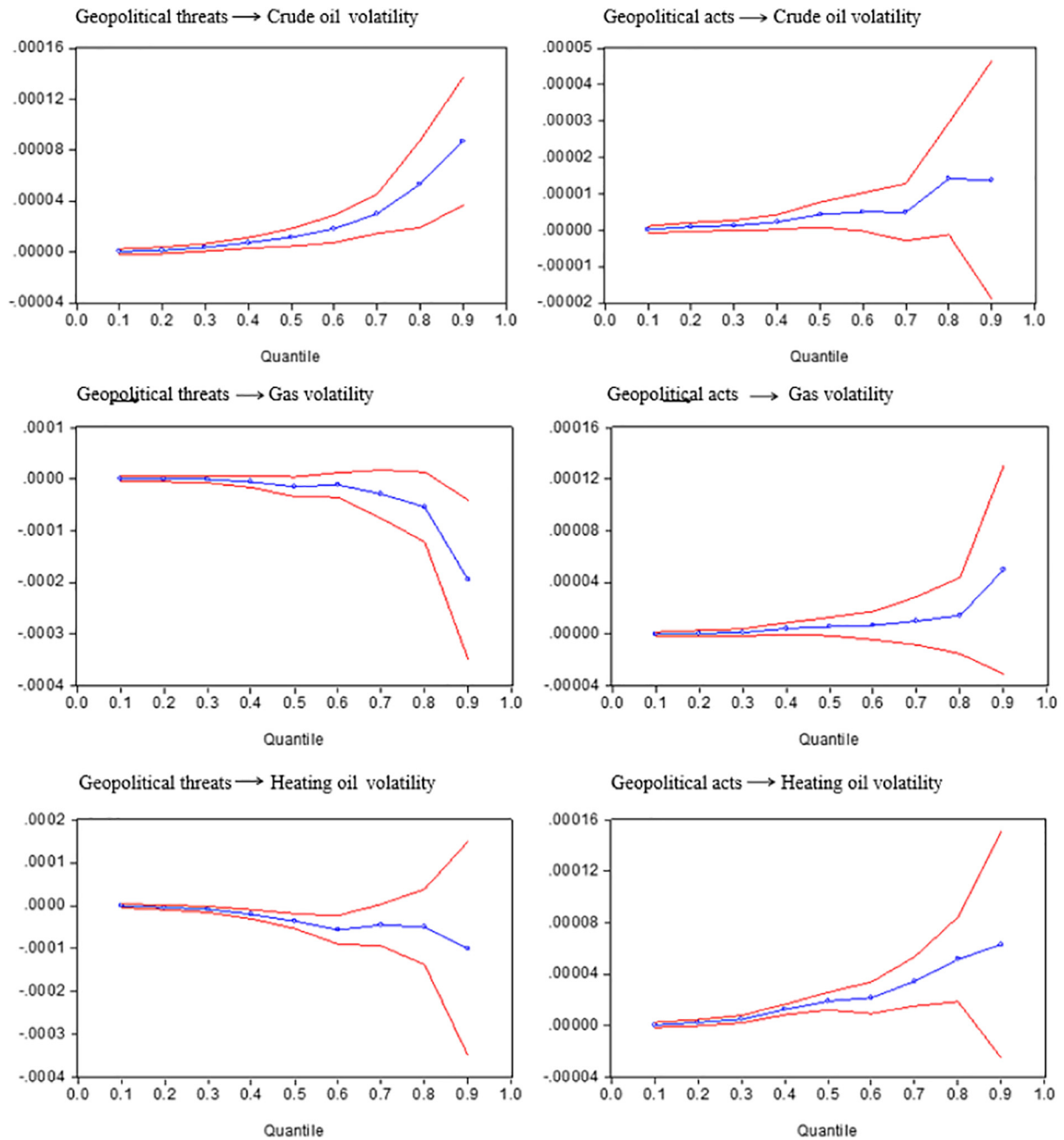


Fig. 2. Quantile regression results with 95% confidence intervals for the effects of geopolitical threats and acts on energy volatility.

Meanwhile, geopolitical threats have significantly positive effects on crude oil volatility in different quantiles but have significantly negative effects on gas and heating oil volatility at various quantiles. Geopolitical acts have significantly positive impacts on energy volatility at different quantiles. These results indicate that geopolitical threats and acts have diverse effects on energy returns and volatility under different market conditions.

Our conclusions have some important implications. First, higher geopolitical risks cause lower crude oil returns and higher crude oil volatility. Geopolitical risks have the ability to predict the levels of crude oil returns and volatility, so the investors need to pay more attention to the prediction ability of geopolitical risks. When geopolitical risks increase, the investors should take corresponding measures to reduce losses in crude oil returns and decrease crude oil

volatility. Second, geopolitical risks have significantly negative impacts on crude oil returns in the bearish market and on heating oil returns in the normal and bullish markets and have significantly positive impacts on crude oil volatility under different market conditions. Therefore, different types of energy respond differently to geopolitical risks under various market conditions. The government departments need to adjust their policies according to the different types of energy and various market conditions to avoid the adverse effects of geopolitical risks. Third, geopolitical threats and acts have diverse effects on energy returns and volatility, and geopolitical risks have various impacts on the returns and volatility of crude oil, gas and heating oil. The policy makers should create different policies based on the two subindexes of geopolitical risks and the three types of energy returns and volatility.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (Nos. 71671187, 71874210, 71633006, 71573282 and 71874207).

References

- Andersen, T.G., Bollerslev, T., 1998. Answering the skeptics: yes, standard volatility models do provide accurate forecasts. *Int. Econ. Rev.* 885–905.
- Antonakakis, N., Gupta, R., Kollias, C., Papadamou, S., 2017. Geopolitical risks and the oil-stock nexus over 1899–2016. *Financ. Res. Lett.* 23, 165–173.
- Asai, M., Gupta, R., McAleer, M., 2020. Forecasting volatility and co-volatility of crude oil and gold futures: effects of leverage, jumps, spillovers, and geopolitical risks. *Int. J. Forecast.* <https://doi.org/10.1016/j.ijforecast.2019.10.003>.
- Baur, D.G., 2013. The structure and degree of dependence: a quantile regression approach. *J. Bank. Financ.* 37, 786–798.
- Bouiyour, J., Selmi, R., Hammoudeh, S., Wohar, M.E., 2019. What are the categories of geopolitical risks that could drive oil prices higher? Acts or threats? *Energy Econ.* 104523.
- Caldara, D., Iacoviello, M., 2016. Measuring geopolitical risk. Board of Governors of the Federal Reserve Board. Working Paper, Version: May 2, 2016.
- Caldara, D., Iacoviello, M., 2018. Available at SSRN: 2018. Measuring geopolitical risk. FRB International Finance Discussion Paper No. 1222. <https://ssrn.com/abstract=3117773>.
- Chen, G.Q., Wu, X.F., 2017. Energy overview for globalized world economy: source, supply chain and sink. *Renew. Sust. Energy Rev.* 69, 735–749.
- Chen, J., Zhu, X., 2019. The effects of different types of oil price shocks on industrial PPI: evidence from 36 sub-industries in China. *Emerg. Mark. Financ. Trade* 11, 1–24.
- Chen, Z.M., Chen, P.L., Ma, Z.M., Xu, S.Y., Hayat, T., Alsaedi, A., 2019. Inflationary and distributional effects of fossil energy price fluctuation on the Chinese economy. *Energy* 187, 115974.
- Chen, J., Zhu, X., Li, H., 2020. The pass-through effects of oil price shocks on China's inflation: a time-varying analysis. *Energy Econ.* 104695.
- Demiralay, S., Kilincarslan, E., 2019. The impact of geopolitical risks on travel and leisure stocks. *Tour. Manag.* 75, 460–476.
- Demirer, R., Gupta, R., Suleman, T., Wohar, M.E., 2018. Time-varying rare disaster risks, oil returns and volatility. *Energy Econ.* 75, 239–248.
- Demirer, R., Gupta, R., Ji, Q., Tiwari, A.K., 2019. Geopolitical risks and the predictability of regional oil returns and volatility. *OPEC Energy Rev.* 43, 342–361.
- Du, L., He, Y., 2015. Extreme risk spillovers between crude oil and stock markets. *Energy Econ.* 51, 455–465.
- Gong, X., Lin, B., 2017. Forecasting the good and bad uncertainties of crude oil prices using a HAR framework. *Energy Econ.* 67, 315–327.
- Gong, X., Lin, B., 2018. The incremental information content of investor fear gauge for volatility forecasting in the crude oil futures market. *Energy Econ.* 74, 370–386.
- Jin, X., 2015. Asymmetry in return and volatility spillover between China's interbank and exchange T-bond markets. *Int. Rev. Econ. Financ.* 37, 340–353.
- Kannadhasan, M., Das, D., 2019. Do Asian emerging stock markets react to international economic policy uncertainty and geopolitical risk alike? A quantile regression approach. *Financ. Res. Lett.* 48, 1–19.
- Koenker, R., Bassett Jr., G., 1978. Regression Quantiles. *Journal of the Econometric Society, Econometrica*, pp. 33–50.
- Li, B.Y., Chang, C.P., Chu, Y., Sui, B., 2020. Oil prices and geopolitical risks: what implications are offered via multi-domain investigations? *Energy Environ.* 31 (3), 492–516.
- Liu, J., Ma, F., Tang, Y., Zhang, Y., 2019. Geopolitical risk and oil volatility: a new insight. *Energy Econ.* 104548.
- Mei, D., Ma, F., Liao, Y., Wang, L., 2020. Geopolitical risk uncertainty and oil future volatility: evidence from MIDAS models. *Energy Econ.* 86, 104624.
- Mosteller, F., Tukey, J.W., 1977. Data analysis and regression: a second course in statistics. Addison-Wesley Series in Behavioral Science: Quantitative Methods.
- Naifar, N., Shahzad, J., Hammoudeh, S.M., 2020. Dynamic nonlinear impacts of oil price returns and financial uncertainties on credit risks of oil-exporting countries. *Energy Econ.* <https://doi.org/10.1016/j.eneco.2020.104747>.
- Nusair, S.A., Olson, D., 2019. The effects of oil price shocks on Asian exchange rates: evidence from quantile regression analysis. *Energy Econ.* 78, 44–63.
- Peng, C., Zhu, H., Guo, Y., Chen, X., 2018. Risk spillover of international crude oil to China's firms: evidence from granger causality across quantile. *Energy Econ.* 72, 188–199.
- Plakandaras, V., Gupta, R., Wong, W.K., 2019. Point and density forecasts of oil returns: the role of geopolitical risks. *Res. Policy* 62, 580–587.
- Reboredo, J.C., Uddin, G.S., 2016. Do financial stress and policy uncertainty have an impact on the energy and metals markets? A quantile regression approach. *Int. Rev. Econ. Financ.* 43, 284–298.
- Saint Akadiri, S., Eluwole, K.K., Akadiri, A.C., Avci, T., 2020. Does causality between geopolitical risk, tourism and economic growth matter? Evidence from Turkey. *J. Hosp. Tour. Manag.* 43, 273–277.
- Su, C.W., Khan, K., Tao, R., Nicoleta-claudia, M., 2019. Does geopolitical risk strengthen or depress oil prices and financial liquidity? Evidence from Saudi Arabia. *Energy* 187, 116003.
- Tiwari, A.K., Das, D., Dutta, A., 2019. Geopolitical risk, economic policy uncertainty and tourist arrivals: evidence from a developing country. *Tour. Manag.* 75, 323–327.
- Wang, W.X., Yang, F.Y., 2018. The shale revolution, geopolitical risk, and oil price volatility. *SSRN Electron. J.* <https://doi.org/10.2139/ssrn.3241692>.
- Wen, F., Xiao, J., Huang, C., Xia, X., 2018. Interaction between oil and US dollar exchange rate: nonlinear causality, time-varying influence and structural breaks in volatility. *Appl. Econ.* 50, 319–334.
- Wen, F., Min, F., Zhang, Y.J., Yang, C., 2019. Crude oil price shocks, monetary policy, and China's economy. *Int. J. Financ. Econ.* 24 (2), 812–827.
- Xiao, J., Zhou, M., Wen, F., Wen, F., 2018. Asymmetric impacts of oil price uncertainty on Chinese stock returns under different market conditions: evidence from oil volatility index. *Energy Econ.* 74, 777–786.
- Xiao, J., Hu, C., Ouyang, G., Wen, F., 2019. Impacts of oil implied volatility shocks on stock implied volatility in China: empirical evidence from a quantile regression approach. *Energy Econ.* 80, 297–309.
- You, W., Guo, Y., Zhu, H., Tang, Y., 2017. Oil price shocks, economic policy uncertainty and industry stock returns in China: asymmetric effects with quantile regression. *Energy Econ.* 68, 1–18.
- Zhu, H., Huang, R., Wang, N., Hau, L., 2019. Does economic policy uncertainty matter for commodity market in China? Evidence from quantile regression. *Appl. Econ.* 1–17.